Tourism Vehicle Booking and Trip Tracking System

Background Analysis and System Design

1. Introduction to the Topic

The tourism industry requires reliable transportation services for activities such as airport transfers, hotel transfers, sightseeing trips, and point-to-point journeys. These services require coordination between customers, vehicles, drivers, bookings, trip schedules, and vehicle maintenance.

When these activities are managed manually using telephone calls, spreadsheets, or separate records, it can be difficult to identify available vehicles, assign suitable drivers, monitor ongoing trips, and maintain accurate historical records.

Therefore, I propose a Tourism Vehicle Booking and Trip Tracking System, which is a web-based system for managing the main activities involved in tourism transportation. Customers can submit transportation requests, while authorized transport staff can review bookings, check vehicle and driver availability, and make assignments. Drivers can manage their assigned trips and share basic location information while a trip is active.

The system will also maintain trip records, vehicle maintenance information and basic reports. A sandbox payment facility will be used to demonstrate the payment process without processing real financial transactions.

2. Background Analysis

2.1 Stakeholder Analysis

Stakeholder	Role / Interest in the System
Tourist / Customer	Requests transportation services, manages bookings, views booking status, makes demo payments and tracks active trips.
Transport Officer / Manager	Manages booking requests, approves/rejects bookings, checks availability and assigns vehicles and drivers.
Driver	Views assigned trips, starts and completes trips, records trip information and provides location information during active trips.
Maintenance Officer	Maintains vehicle service records and identifies vehicles that require maintenance.
Sandbox Payment Gateway	Supports the demonstration of the booking payment process.
Browser Geolocation API	Provides the driver's location during an active trip.

3. System Requirements

3.1 Functional Requirements

User Authentication

- FR01: The system shall allow customers to register for an account.
- FR02: The system shall allow registered users to log in.
- FR03: The system shall provide role-based access to system functions.

Vehicle Booking

- FR04: The system shall allow customers to submit transportation booking requests.
- FR05: The system shall allow customers to enter pickup location, destination, date, time, passenger count and trip type.
- FR06: The system shall allow customers to view their booking status.
- FR07: The system shall allow transport staff to view submitted booking requests.
- FR08: The system shall allow transport staff to approve or reject booking requests.

Vehicle and Driver Assignment

- FR09: The system shall allow transport staff to check vehicle availability.
- FR10: The system shall allow transport staff to check driver availability.
- FR11: The system shall allow transport staff to assign a suitable vehicle to an approved booking.
- FR12: The system shall allow transport staff to assign a suitable driver to an approved booking.

Trip Management

- FR13: The system shall allow drivers to view their assigned trips.
- FR14: The system shall allow drivers to start an assigned trip.
- FR15: The system shall allow drivers to record the starting odometer reading.
- FR16: The system shall allow drivers to update the trip status.
- FR17: The system shall allow drivers to provide their current location during an active trip.
- FR18: The system shall allow drivers to complete a trip.
- FR19: The system shall allow drivers to record the ending odometer reading.
- FR20: The system shall allow drivers to add remarks to completed trips.

Trip Tracking

- FR21: The system shall allow customers to track an active trip.
- FR22: The system shall allow authorized transport staff to monitor active trips.
- FR23: The system shall store basic current or last-known location information for active trips.

Payment

- FR24: The system shall provide a sandbox/demo payment process for bookings.
- FR25: The system shall record the payment status of a demonstration payment.

Vehicle and Driver Management

- FR26: The system shall allow authorized staff to add, update, view and manage vehicle information.
- FR27: The system shall allow authorized staff to add, update, view and manage driver information.

Vehicle Maintenance

- FR28: The system shall allow authorized users to add, update and view vehicle maintenance records.
- FR29: The system shall allow users to record service dates and mileage.
- FR30: The system shall allow users to record the next service requirement.
- FR31: The system shall allow users to view vehicle maintenance history.
- FR32: The system shall identify vehicles that are due for maintenance or temporarily unavailable.

Reports and History

- FR33: The system shall maintain booking and trip history.
- FR34: The system shall provide basic reports related to bookings, trips, vehicles and maintenance.

3.2 Non-Functional Requirements

NFR01 – Usability: The system should provide a simple and understandable interface so that each user can perform their relevant tasks easily.

NFR02 – Performance: The system should respond to normal user requests within a reasonable time under the expected project workload.

NFR03 – Security: The system should restrict access to functions according to the user's role and require authentication for protected functions.

NFR04 – Reliability: The system should store booking, trip, vehicle and maintenance information accurately and consistently.

NFR05 – Data Persistence: The system should retain stored information in the MySQL database when users log out or the application is restarted.

NFR06 – Maintainability: The application should be organized into understandable components so that errors can be identified and changes can be made more easily.

NFR07 – Browser Compatibility: The system should be accessible through commonly used modern web browsers.

NFR08 – Responsiveness: The user interface should be usable on different screen sizes through the responsive features provided by Bootstrap.

4. Use Case Modelling

4.1 Identified Actors

- Tourist / Customer
- Transport Officer / Manager
- Driver
- Maintenance Officer
- Sandbox Payment Gateway
- Browser Geolocation API

4.2 Main Use Cases

Tourist / Customer

- Register
- Login
- Make Vehicle Booking
- View Booking Status
- View Assigned Vehicle and Driver
- Make Demo Payment
- Track Active Trip
- View Booking/Trip History

Transport Officer / Manager

- Manage Booking Requests
- Approve/Reject Booking
- Check Vehicle Availability
- Check Driver Availability
- Assign Vehicle
- Assign Driver
- Monitor Active Trips
- Manage Vehicles
- Manage Drivers
- View Reports

Driver

- Login
- View Assigned Trips
- Start Trip
- Record Starting Odometer
- Update Trip Status
- Share Location
- Complete Trip
- Record Ending Odometer
- Add Trip Remarks

Maintenance Officer

- Manage Maintenance Records
- View Maintenance History
- Identify Vehicles Due for Maintenance

External Actors

- Sandbox Payment Gateway – Process Demo Payment
- Browser Geolocation API – Provide Current Location

4.3 Use Case Diagram

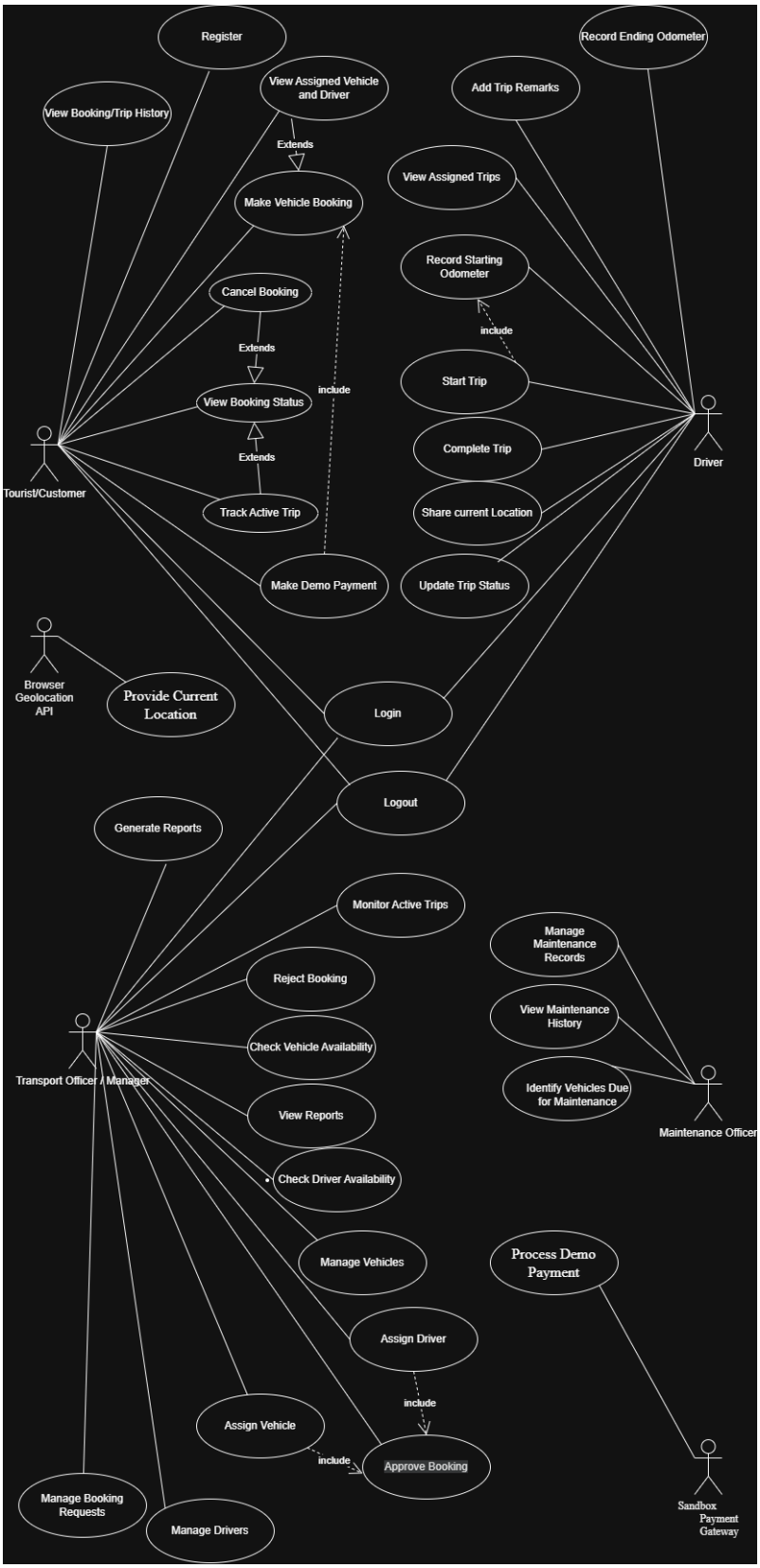


Figure 1 : Use Case Diagram of the Tourism Vehicle Booking and Trip Tracking System

4.4 Use Case Narratives

UC01 – Make Vehicle Booking

Use Case ID	UC01
Use Case Name	Make Vehicle Booking
Primary Actor	Tourist / Customer
Precondition	Customer has logged into the system.
Main Flow	1. Customer selects the vehicle booking option. 2. System displays the booking form. 3. Customer enters pickup location, destination, date, time, passenger count and trip type. 4. Customer submits the booking. 5. System validates the information. 6. System creates the booking with Pending status.
Alternative Flow	If required information is missing or invalid, the system displays an error and asks the customer to correct the information.
Postcondition	The booking request is stored and available for transport staff to review.

UC02 – Approve and Assign Vehicle/Driver

Use Case ID	UC02
Use Case Name	Approve and Assign Vehicle/Driver
Primary Actor	Transport Officer / Manager
Precondition	A customer booking has been submitted.
Main Flow	1. Officer views booking requests. 2. Officer selects a booking. 3. Officer reviews booking details. 4. Officer approves the booking. 5. System displays available vehicles and drivers. 6. Officer selects a suitable vehicle and driver. 7. Officer confirms the assignment. 8. System updates the booking status to Assigned.
Alternative Flow	If the booking is not suitable, the officer can reject it. If no suitable vehicle or driver is available, the assignment cannot be completed.
Postcondition	The approved booking has an assigned vehicle and driver.

UC03 – Manage Trip

Use Case ID	UC03
Use Case Name	Manage Trip
Primary Actor	Driver
Precondition	A vehicle and driver have been assigned to the booking.
Main Flow	1. Driver logs into the system. 2. Driver views assigned trip. 3. Driver starts the trip. 4. Driver records the starting odometer. 5. System changes the trip status to In Progress. 6. Driver updates trip information during the journey. 7. Driver records the ending odometer. 8. Driver adds remarks if necessary. 9. Driver completes the trip. 10. System changes the status to Completed.
Alternative Flow	If the driver cannot complete the trip, the relevant status or remark can be recorded.
Postcondition	The completed trip and relevant trip information are stored.

UC04 – Track Active Trip

Use Case ID	UC04
Use Case Name	Track Active Trip
Primary Actor	Tourist / Customer
Precondition	The assigned trip is currently in progress.
Main Flow	1. Driver starts the trip. 2. The browser obtains the driver's location. 3. The system receives the location information. 4. The location is stored. 5. Customer views the current/last-known location of the trip.
Alternative Flow	If location information is temporarily unavailable, the system can retain the last-known location.
Postcondition	Basic location information for the active trip is available.

5. System Design

5.1 High-Level System Architecture

The proposed system can be represented using simple three-layer web architecture. The presentation layer provides the user interface, the application layer contains the PHP-based business logic, and the data layer uses MySQL for persistent storage. External services include the sandbox payment gateway and browser geolocation API.

Presentation Layer: HTML5, CSS3, Bootstrap 5 and JavaScript.
Application Layer: PHP 8.2 for authentication, booking, assignment, trip management, tracking, maintenance and reports.
Data Layer: MySQL 8.0 for persistent system data.
External Services: Sandbox Payment Gateway and Browser Geolocation API.

5.2 Detailed System Design

5.2.1 Main Classes

User • userID • name • email • passwordHash • role • phoneNumber • accountStatus • createdAt	**Customer** • customerID • userID • country • address	**Driver** • driverID • userID • NIC • address • licenseNumber • licenseExpiryDate • availabilityStatus	**TransportOfficer** • officerID • userID
MaintenanceOfficer • maintenanceOfficerID • userID	**Vehicle** • vehicleID • registrationNumber • make • model • year • capacity • currentOdometer • status	**Booking** • bookingID • customerID • pickupLocation • destination • bookingDate • bookingTime • passengerCount • tripType • specialRequests • totalAmount • status • createdAt	**Trip** • tripID • bookingID • vehicleID • driverID • tripStatus • startTime • endTime • startOdometer • endOdometer • distance • remarks

LocationRecord	Payment	MaintenanceRecord	
• locationID • tripID • latitude • longitude • recordedTime	• paymentID • bookingID • amount • paymentReference • paymentMethod • paymentStatus • paymentDate	• maintenanceID • vehicleID • maintenanceType • serviceDate • mileage • nextServiceDate • nextServiceMileage • remarks	

Table 1 : Main Classes

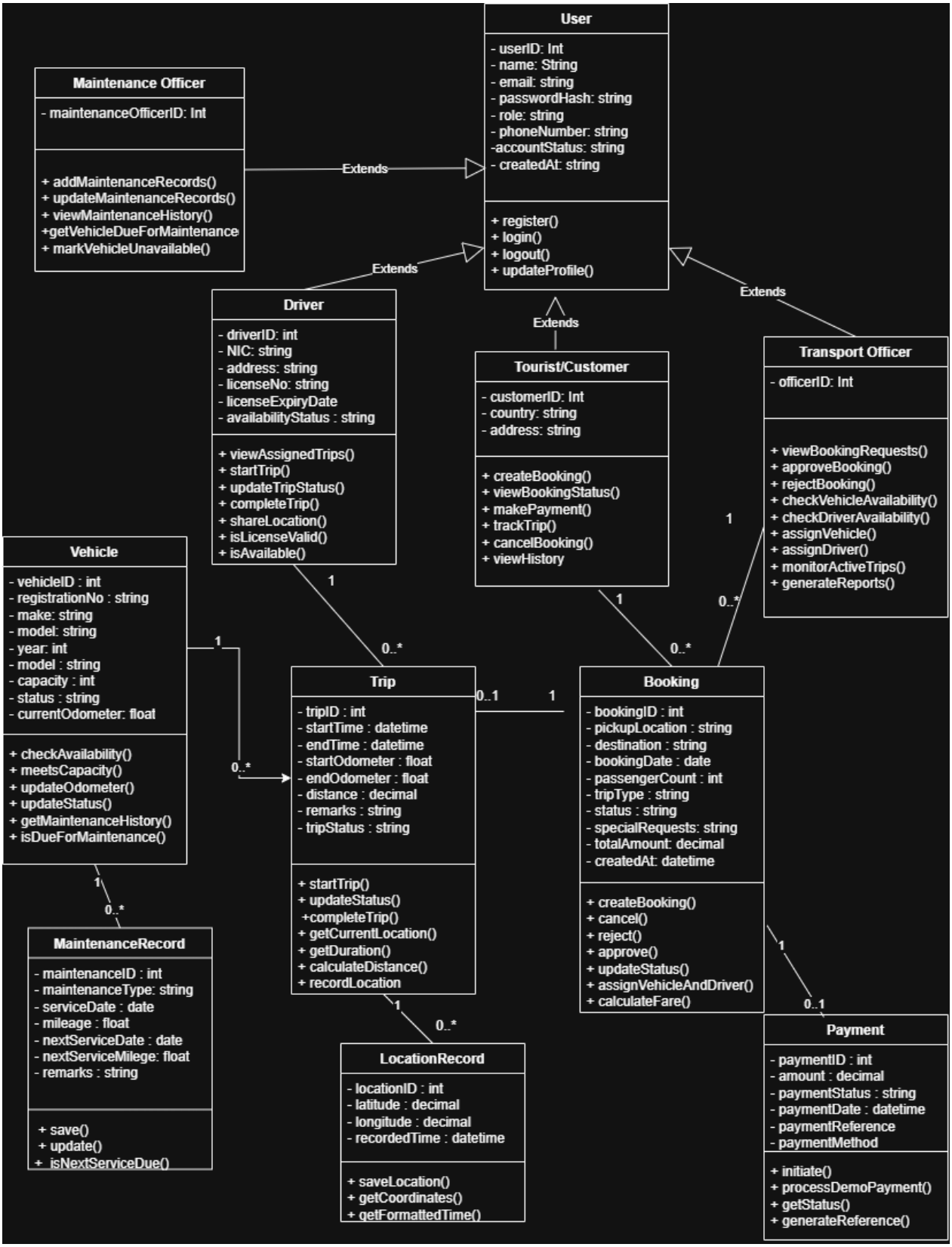


Figure 2: Class Diagram

5.2.2 State Diagram

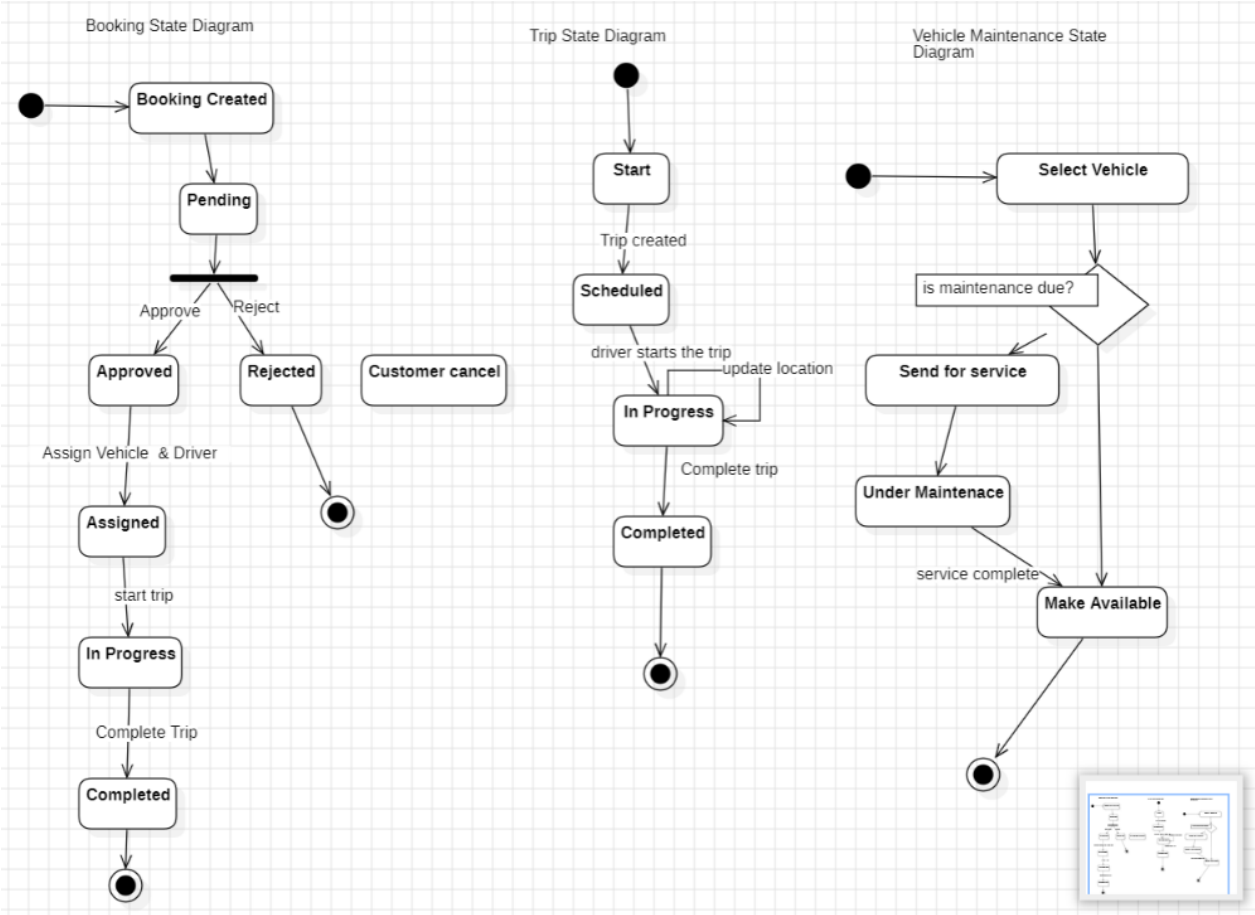


Figure 3 : State Diagrams

5.3 Database Design

5.3.1 Main Entities

- User
- Customer
- Driver
- Transport Officer
- Maintenance Officer
- Vehicle
- Booking
- Trip
- Location Record
- Payment
- Maintenance Record

5.3.2 Main Relationships

- A Customer can have multiple Bookings.
- A Booking can have a Trip.
- A Trip is associated with one assigned Vehicle.
- A Trip is associated with one assigned Driver.
- A Trip can have multiple Location Records.
- A Booking can have a Payment record for the demonstration payment.
- A Vehicle can have multiple Maintenance Records.

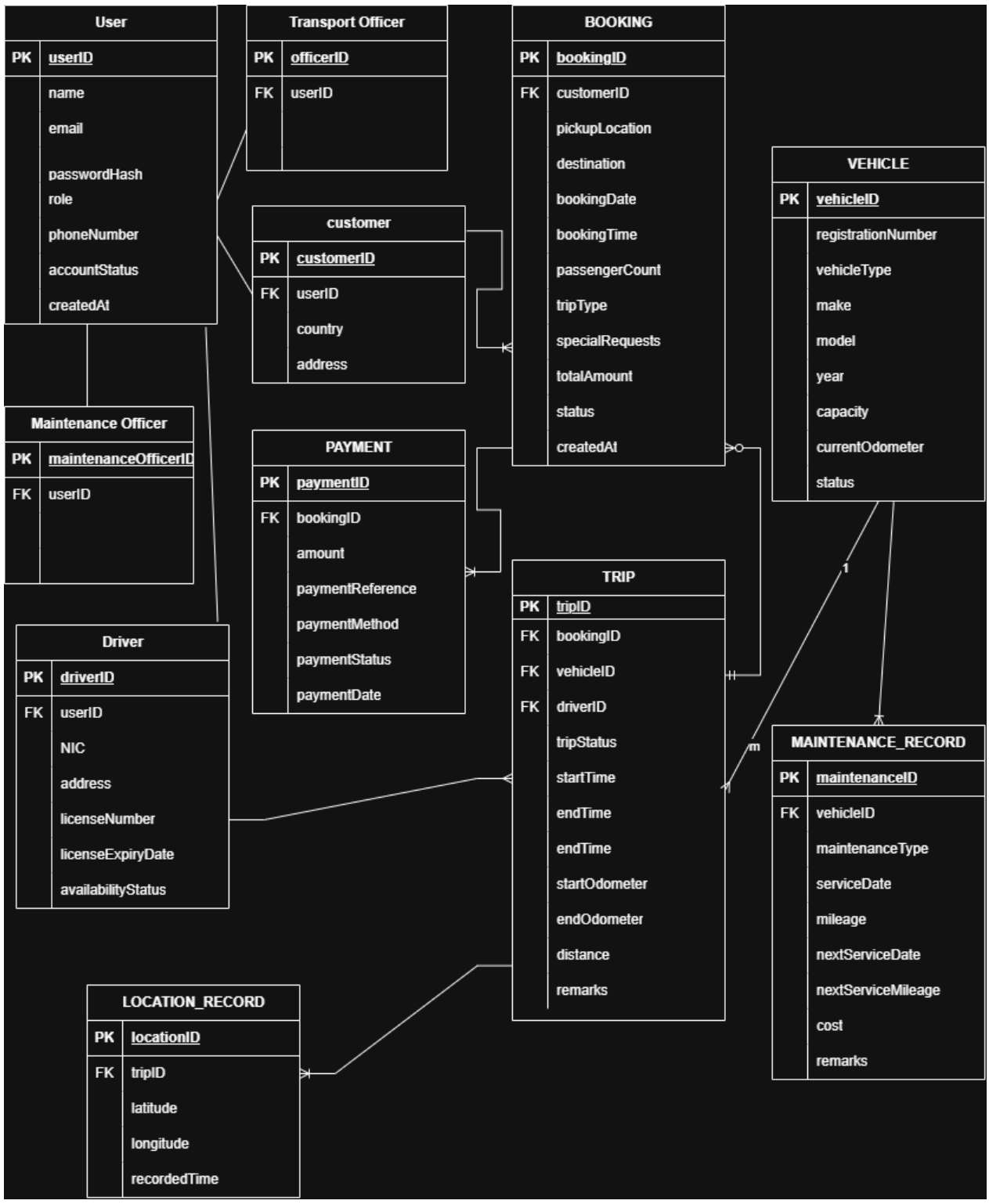


Figure 4: ER Diagram

5.4 User Interface Design – Wireframes

The wireframes will focus on the main workflows rather than every possible screen.

Customer

- Login Page
- Customer Dashboard
- Vehicle Booking Form
- My Bookings / Booking Details
- Active Trip Tracking

Transport Officer

- Officer Dashboard
- Booking Management
- Vehicle and Driver Assignment
- Active Trip Monitoring
- Reports

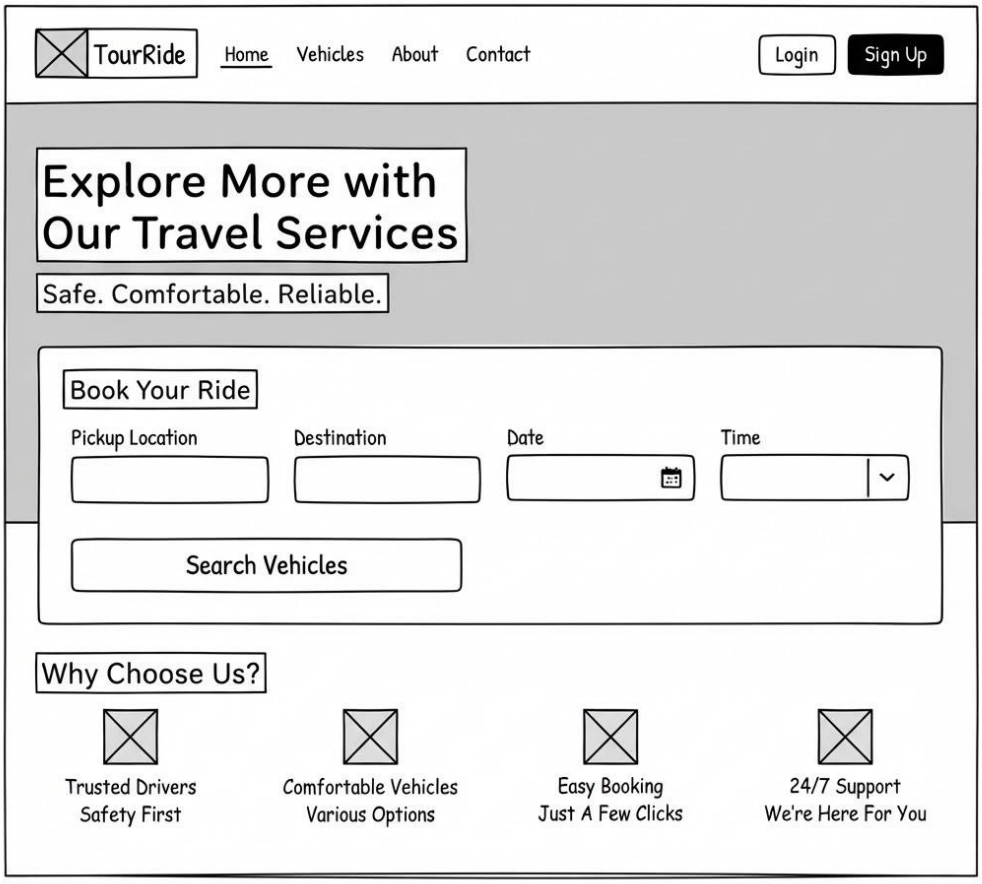
Driver

- Driver Dashboard
- Assigned Trip
- Start/Active Trip
- Complete Trip

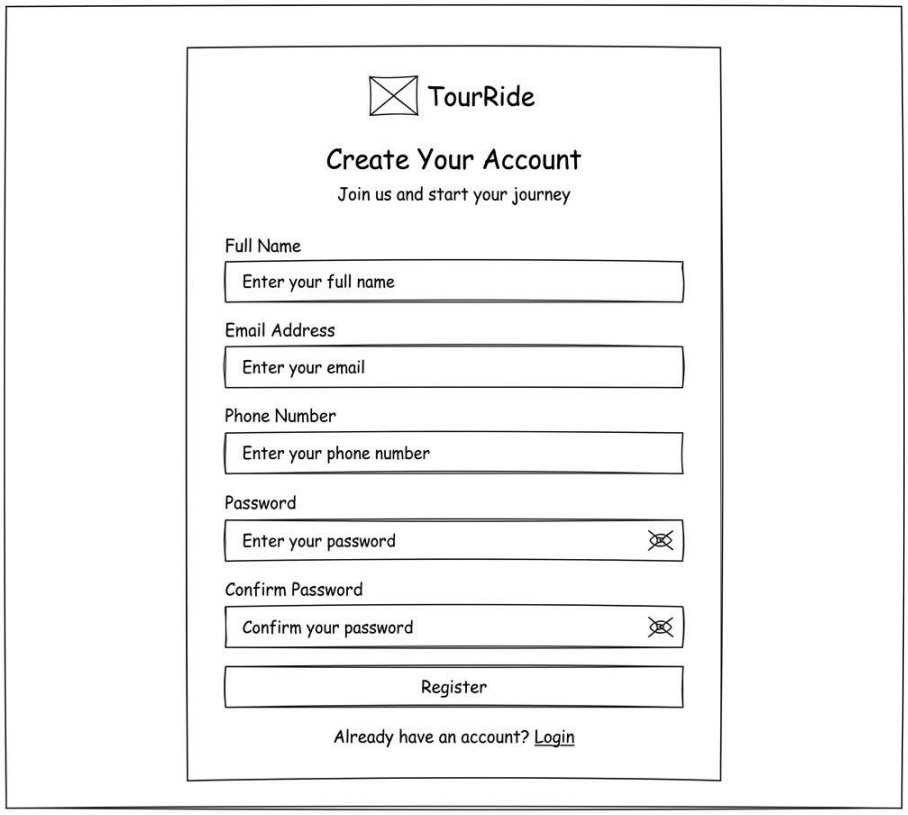
Maintenance Officer

- Maintenance Dashboard
- Vehicle Maintenance Records
- Add Maintenance Record
- Maintenance Due

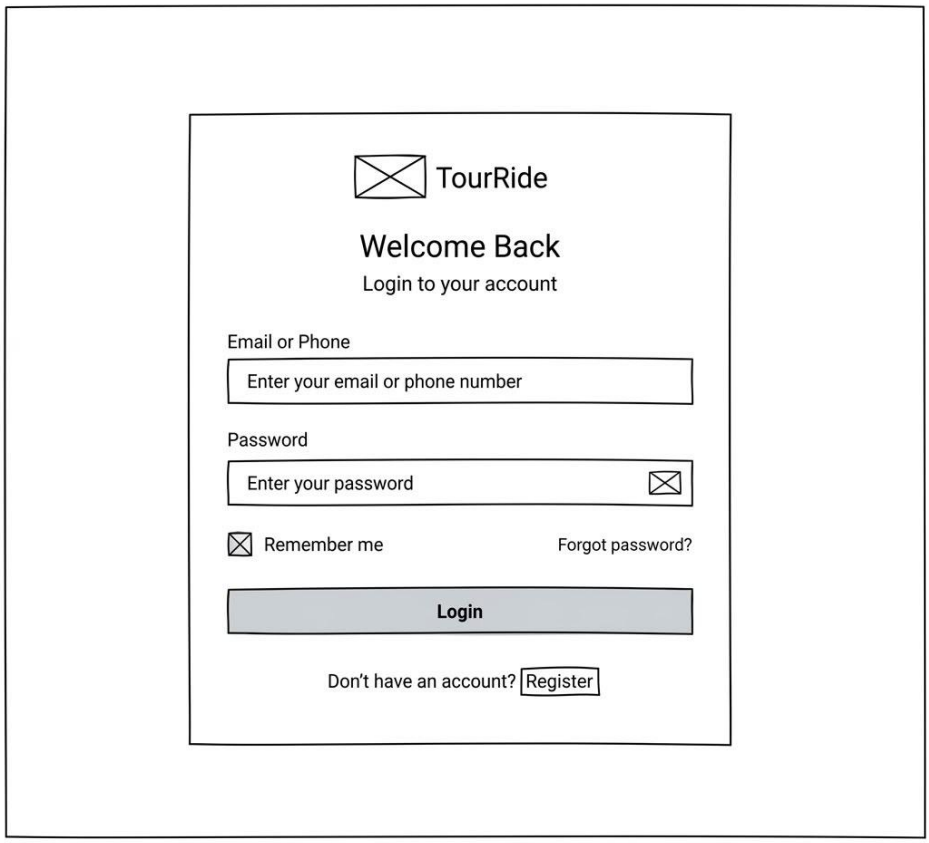
1. Homepage (Public)



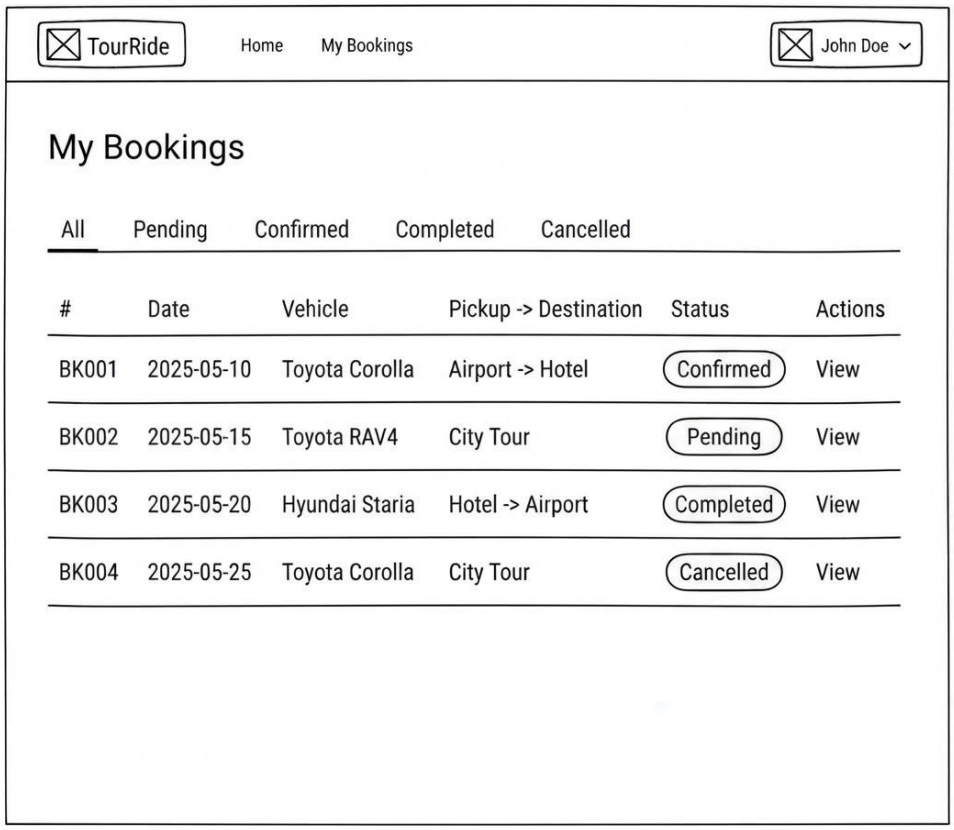
2. User Registration (Customer)



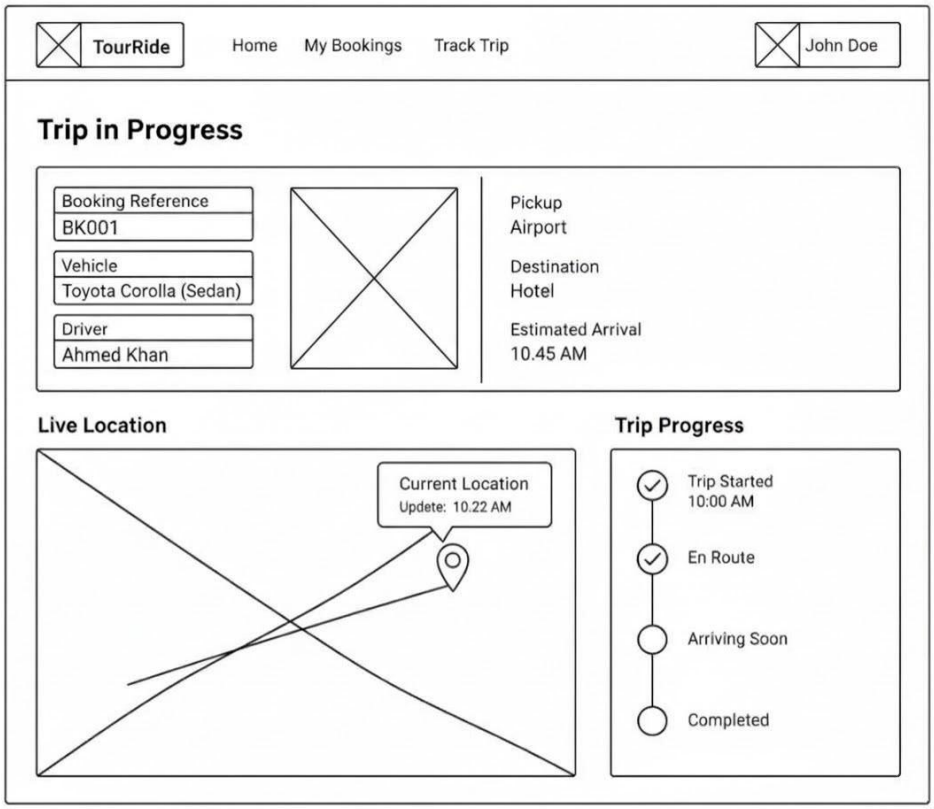
3. Login



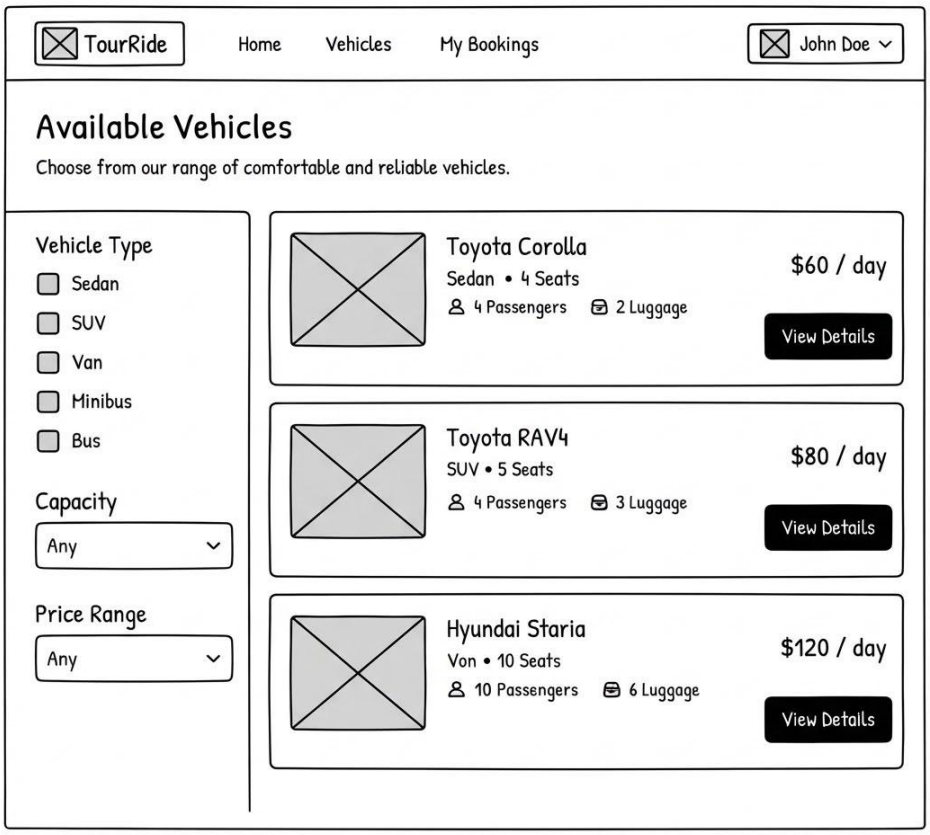
7. My Bookings (Customer)



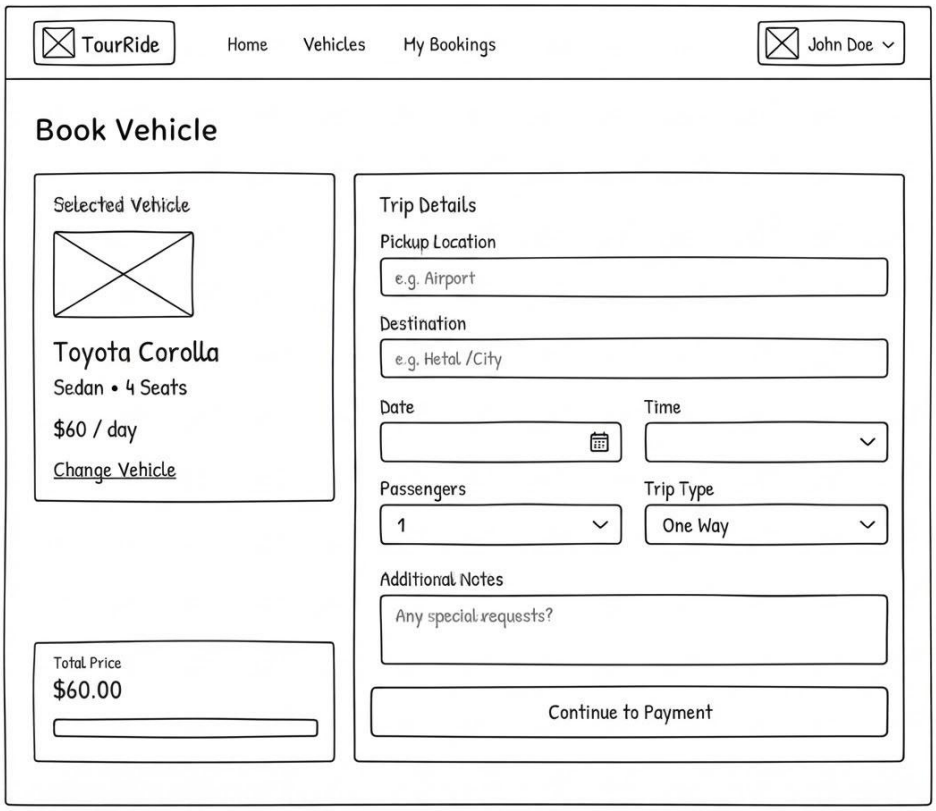
8. Live Trip Tracking (Customer)



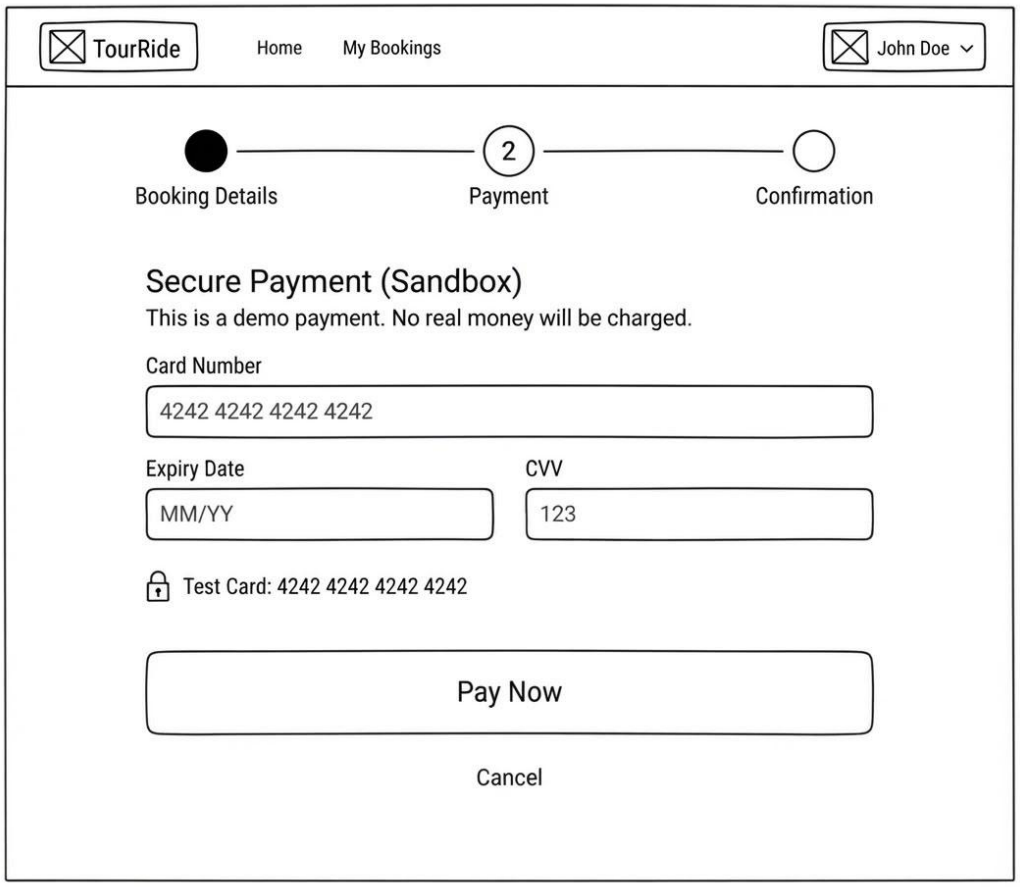
4. Vehicle List / Browse (Customer)



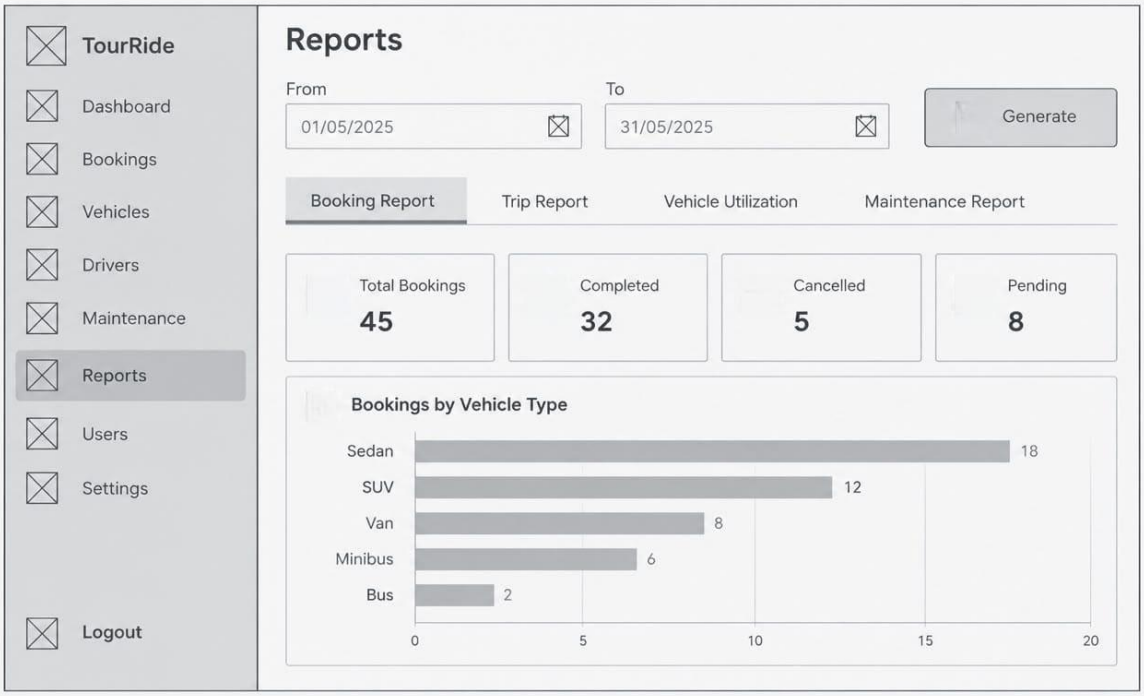
5. Booking Form (Customer)

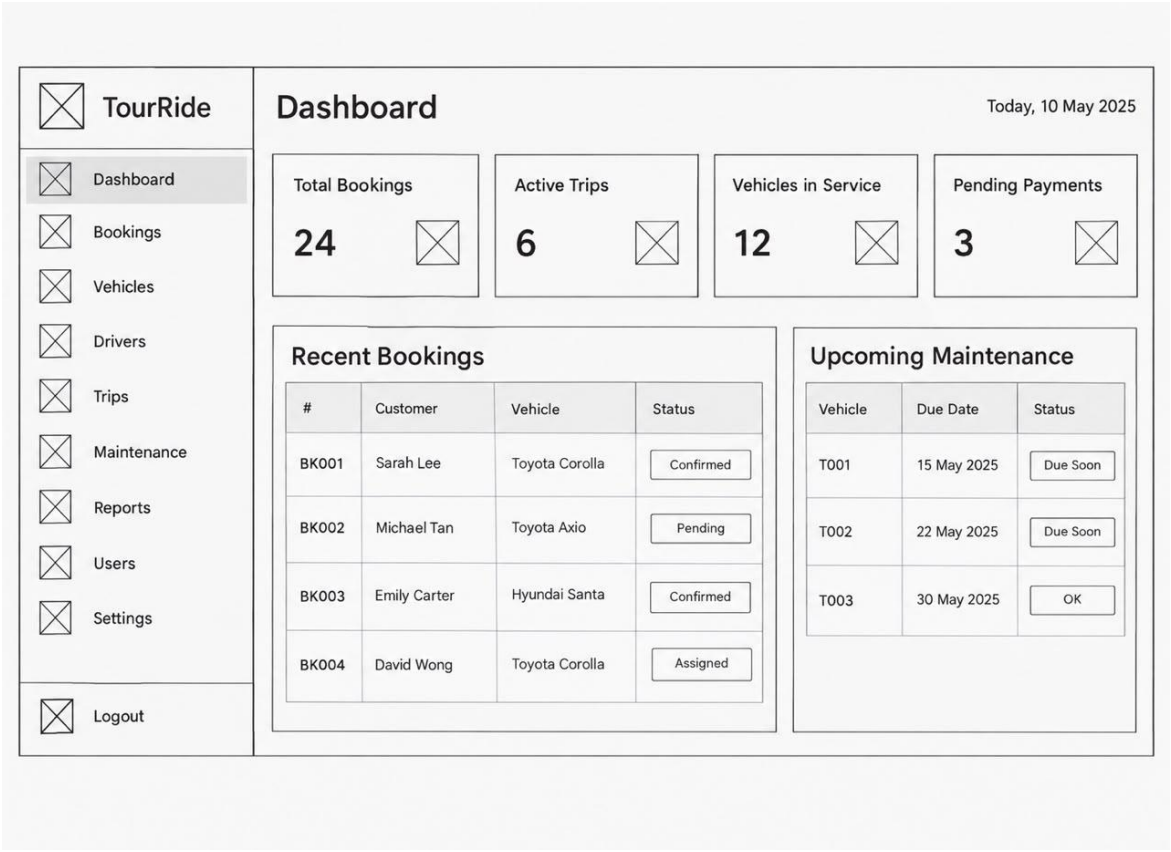


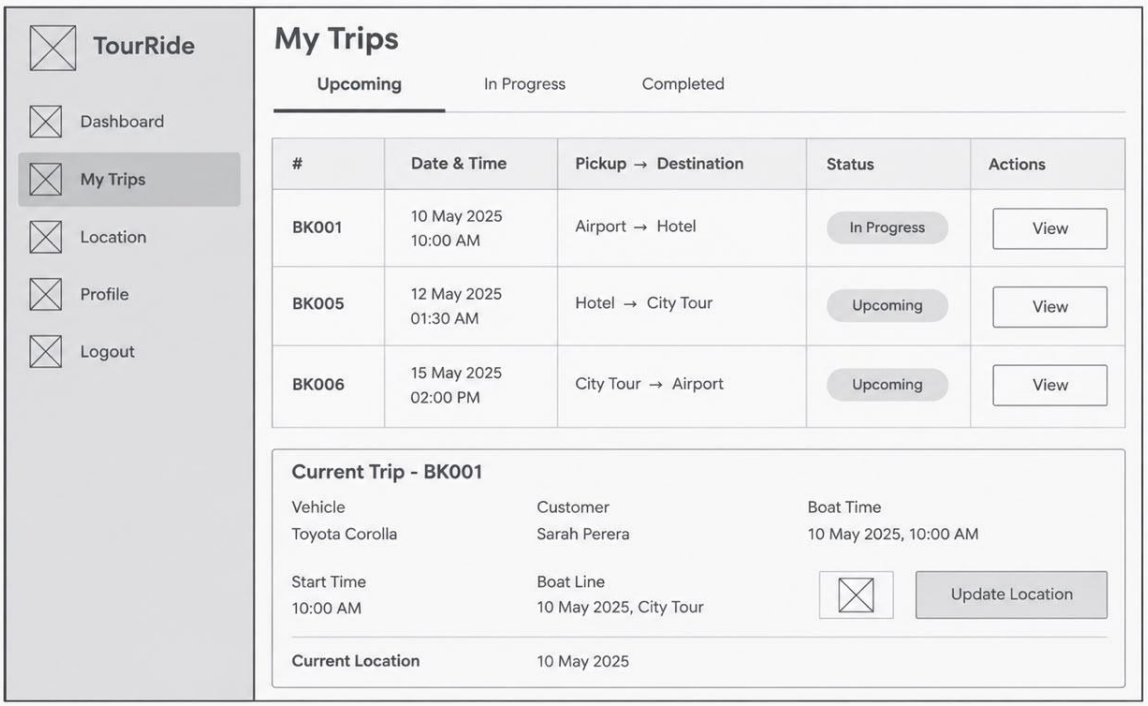
6. Payment (Sandbox)

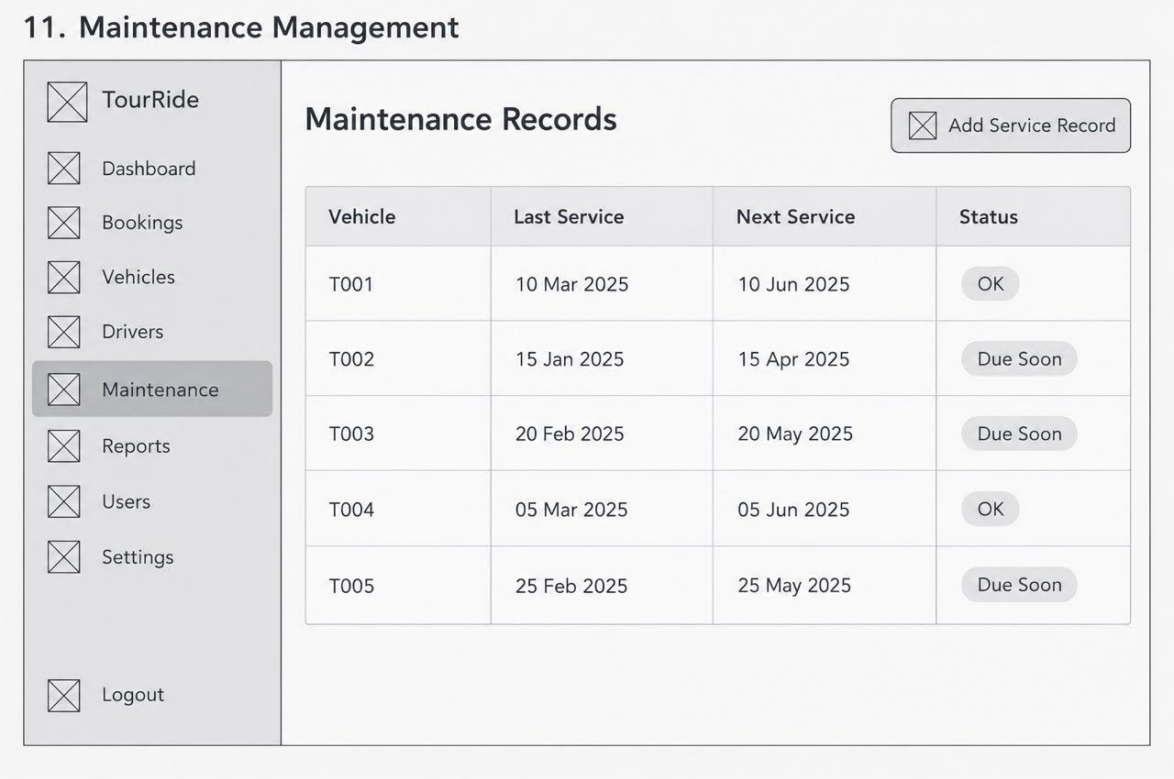


12. Reports




TourRide
Dashboard
Bookings
Vehicles
Drivers
Trips
Maintenance
Reports
Users
Settings
Logout
Dashboard
Today, 10 May 2025
Total Bookings
24
Active Trips
6
Vehicles in Service
12
Pending Payments
3
Recent Bookings
Customer Vehicle Status
BK001 Sarah Lee Toyota Corolla Confirmed
BK002 Michael Tan Toyota Axio Pending
BK003 Emily Carter Hyundai Santa Confirmed
BK004 David Wong Toyota Corolla Assigned
Upcoming Maintenance
Vehicle Due Date Status
T001 15 May 2025 Due Soon
T002 22 May 2025 Due Soon
T003 30 May 2025 OK


TourRide
Dashboard
My Trips
Location
Profile
Logout
My Trips
Upcoming
In Progress
Completed
Date & Time Pickup → Destination Status Actions
BK001 10 May 2025 10:00 AM Airport → Hotel In Progress View
BK005 12 May 2025 01:30 AM Hotel → City Tour Upcoming View
BK006 15 May 2025 02:00 PM City Tour → Airport Upcoming View
Current Trip - BK001
Vehicle
Toyota Corolla
Customer
Sarah Perera
Boat Time
10 May 2025, 10:00 AM
Start Time
10:00 AM
Boat Line
10 May 2025, City Tour
Update Location
Current Location
10 May 2025



5.5. Business Rules

BR01 – Vehicle Capacity

A vehicle shall only be assigned when its passenger capacity is equal to or greater than the booking passenger count.

BR02 – Vehicle Availability

A vehicle that is already assigned to another overlapping trip or is under maintenance shall not be assigned.

BR03 – Driver Availability

A driver shall not be assigned to overlapping trips.

BR04 – Driver Licence

A driver with an expired driving licence shall not be assigned to a trip.

BR05 – Odometer

The ending odometer reading shall not be lower than the starting odometer reading.

BR06 – Vehicle Odometer

A vehicle's current odometer value shall not decrease.

BR07 – Trip Completion

A trip shall not be completed unless a valid starting odometer reading has been recorded.

BR08 – Location Tracking

Location tracking shall only be active while the trip is in progress.

BR09 – Maintenance

A vehicle identified as unavailable due to maintenance shall not be assigned to a new trip.

BR10 – Demo Payment

Only sandbox/demo payment transactions shall be processed; no real financial transaction shall be performed.

Index No- 24550825

Name – K.A.D.G.T.Wijepala